Lab 18: Evolution — Tragedy of the Commons and Game Theory (Fishing Simulation)

BIOL-8

Name: ____________________ **Date:** ____________________

Group: ____________________

Learning Objectives

By the end of this lab, you will be able to:

- **Explain** the tragedy of the commons in your own words and relate it to a **shared** fish pool your group managed.
- **Describe** how **rotating** who fishes each season and **rising point values** (each season pays more per fish) can change incentives from one round to the next.
- **Apply** the reproduction rule to update population sizes **per species** after a season.
- **Connect** the simulation to **game theory** terms where appropriate: strategy, payoff, cooperation vs. short-term gain, and effects of **repeated** interaction across seasons.
- **Compare** the **four species** by value: gold vs. silver and large vs. small.

Purpose

You will work in a **group** to fish a **shared stock** of four kinds of fish over **three seasons.**

The activity connects **evolution and ecology** to **human behavior** in two ways:

- **Tragedy of the commons** — a shared resource can be depleted when many people pursue short-term private gain.
- **Game theory** — your payoff depends on others' choices; **repeated** seasons change what looks like a good strategy.

After each season, fish that were **not** caught can **reproduce** using a fixed rule. The pool can **recover** or **collapse**, depending on what the group harvests.

Background

Tragedy of the commons

When a **resource** (pasture, fishery, clean air) is **open to many users** and no one person pays the full **future** cost of overuse, each user can be tempted to take as much as possible **now**. If everyone does that, the **shared stock** shrinks and may **collapse**—even though **no one** wanted that outcome. **Rules**, **quotas**, **catch limits**, and **community agreements** exist partly to avoid that outcome. This lab is a **toy** version of that tension.

Game theory in one paragraph

Game theory models choices when your outcome depends on **others'** choices. In a **repeated** setting (here, **three seasons**), what pays off in season 1 can differ from what pays off after others respond in seasons 2 and 3. You may see **defection** (take a lot) vs. **cooperation** (leave fish for the group and the future) or a mix, depending on your group's **norms** and the **rising point values** in the schedule.

The four species and value (relative prices)

Relative value (highest to lowest in this model):

Species	Value in the model
Large goldfish	**Highest** value per fish (gold and large)
Small goldfish	High, but **less** than large gold (still gold)
Large silverfish	**Less** than either gold (silver), but **large** > small for silver
Small silverfish	**Lowest** value per fish

Gold types are worth more than **silver** types. **Large** types are worth more than **small** types, within the same "metal" group.

Prices and seasons: Earnings are in **class points** per fish, using the **fixed schedule** in the **Data** section below (**Table 2**), unless your instructor posts a different table. **Every** species earns **more points** in **season 2** than in **season 1**, and more again in **season 3**—so the pull to overharvest can grow each round if your group does not protect the stock.

Reproduction rule (after each season, before the next)

Work **separately for each of the four species:**

1. Count how many fish of that species were **not** caught (the fish **left in the pool** after that season's harvest). Call this **N** (a whole number).

2. For that species, the pool **gains** new fish. **Divide N by 2 and round down** to a whole number (in math this is the *floor* of N/2). That number is how many **new** fish of that species you **add**.
3. Example: N = 5 → 5 / 2 = 2.5 → **2** new fish (round down). New total = 5 + 2 = **7** before the next season.
4. The **caught** fish are **removed** from the pool. They do **not** count toward reproduction in this model.

If your instructor uses a **different** reproduction rule, **follow their version**.

Connection to "evolution" in BIOL-8

Evolution is not only long-term **genetic** change. Biologists also study how **ecological and social** conditions **select** for behaviors, norms, and structures (e.g. cooperation, punishment of cheaters) that affect who survives and **reproduces**—including in **H. sapiens**. This lab is a **classroom analogy**: no real evolution in the fish, but real discussion about **incentives**, **group outcomes**, and **sustainability**—concepts that sit next to **natural selection** in a broader life-science view.

Materials

- **Shared pool** of countable units for four species (e.g. colored chips, small cards, or tallies) — **large gold**, **small gold**, **large silver**, **small silver**
- A way to record **start**, **catch**, **left**, **reproduction add**, and **end** counts **per species, per season**
- The **point schedule** printed in this handout (Table 2) — *your instructor may substitute; use their values if so*
- This handout and writing tools

Procedure

Before season 1

1. **Form groups** and write **all members' names** in the roster below. You will use **three fishing seasons.**
2. For **each season, one** group member is the **fisher** for that season (someone who was not the fisher the previous time). **Rotate** so **three different people** fish in seasons 1–3 (if the group has fewer than three people, your instructor will say how to handle rotation).

3. Copy the **starting population** for each **species** from the instructor (or the board) into **Table 1** under **Start (Season 1).**
4. Use the **price schedule (points per fish)** printed in **Table 2** below for all earnings, **unless** your instructor gives a different table—then use theirs for every season.

Each season (repeat for seasons 1, 2, and 3)

1. The **fisher** for that season “catches” by **taking** units from the pool as allowed by the rules your instructor gives (e.g. **maximum** number of fish, or a **time** limit, or a **fixed** haul). Record **only** the **number caught per species** in **Table 1** in the **Catch** row for that season.
2. **After** the catch, compute **left** in the pool: **Start − catch** (per species) for that season. Enter under **Left after catch.**
3. **Reproduction:** For **each** species, compute **add** = (**Left after catch**) / 2, **rounded down** to a whole number. New count before the next season = **Left after catch** + **add**. Enter under **Repro add** and **End (after repro).** The **End** row is the **Start** for the next season.
4. **Earnings for this season:** For each species, **(catch)** × **(points for that species in this season)** from the **Table 2** price grid. Add the four products to get **group points** for the season. Do the same for seasons 2 and 3 using the **points columns** for that season in the same table.
5. The **next** group member **fishes** the following season, starting from the new **Start** row.

After season 3

- Total **cumulative** earnings if your instructor requests it.
- Complete **Table 1** and **Table 2**, then the **Analysis** and **Conclusion** sections.

Data

Abbreviations: LG = large gold, SG = small gold, LS = large silver, SS = small silver.

Show at least one season's earnings as a sum of (catch × points) for each species (in Table 2 or the work space below the price table).

Group roster (fill in before Season 1)

Group roster

Role	Name
Fisher — Season 1	
Fisher — Season 2	
Fisher — Season 3	
Other group members (if any)	

Table 1 — Population and catch (all three seasons)

Starting populations (from instructor) go in **Start (S1)** in the Season 1 table. **Left** = after catch; **Repro add** and **End** follow the procedure (reproduction rule).

Table 1 — Season 1

	LG	SG	LS	SS
Start (S1)				
Catch				
Left				
Repro add				
End				

Season 2 — *Fisher (name):*

Table 1 — Season 2

	LG	SG	LS	SS
Start (= S1 end)				
Catch				
Left				
Repro add				
End				

Season 3 — *Fisher (name):*

Table 1 — Season 3

	LG	SG	LS	SS
Start (= S2 end)				
Catch				
Left				
Repro add				
End				

Table 2 — Price schedule (class default) and earnings

Points per fish caught — use these values for **earnings** unless your instructor replaces them. **Gold** rows earn more than **silver**; **large** more than **small**; each **season** pays more per fish than the last.

Species	Points — Season 1	Points — Season 2	Points — Season 3
LG (large gold)	8	12	18
SG (small gold)	4	6	9
LS (large silver)	2	3	5
SS (small silver)	1	2	3

Earnings for a season = **sum** over the four species of **(number caught)** × **(points for that species in that season).**

> **Work space (one season, optional check):** Pick **one** season and write **(catch × points)** for each species, then the **sum** (should match the row you enter in the Earnings table below).

Table 2 — Earnings by season (points)

Season	Group earnings (show work or subtotal)
1	
2	
3	
Total (all seasons)	

Analysis

Answer in **complete sentences**. Use your **tables** for any question that refers to your group's numbers.

Warm-up (introductory). Orient yourself to the simulation before the numbered prompts.

A. What exactly was **shared** by the whole group (the **commons**), and what could each fisher try to **take** during their season?

B. List the four species abbreviations (**LG, SG, LS, SS**) and, in one sentence, how **gold vs. silver** and **large vs. small** relate to **point value** in this model.

C. After a season, fish that were **not** caught could contribute to **reproduction** before the next season. Why does leaving fish in the pool matter for whether the stock can **recover** instead of **collapse**?

1. Tragedy of the commons. In two or three sentences, did your group's pool **trend toward depletion, stability, or recovery** for one or more species? Name **one** choice that made that outcome more likely.

2. Evolution of incentives. Prices rose each season. How did that affect what you think the “best” **short-term** move was, versus a **long-term** group goal?

3. Game theory. Name **one** behavior that acted like **cooperation** and **one** that acted like **defection** (or self-interest only), whether or not you used those words during the game.

4. Reproduction rule. For **one** species, show with numbers: **left after catch**, **repro add** (left / 2, **round down**), and **end** after you add. Did the rule ever **increase** the stock enough that your group was glad some fish were **left in the pool**?

Conclusion

Introductory. One look back at your **numbers** before the final synthesis.

D. From **Table 1** or **Table 2**, name **one specific figure** (a count or an earnings value) that shows whether your group leaned toward **sustainability** or **heavy harvest**—and say why you picked it.

5. Human biology in one sentence: Why might a **lecture** on evolution in humans include a **resource** and **incentive** story like this, not only DNA and fossils?

Lab 18 — Evolution. Tragedy of the commons and game theory via a three-season fishing simulation.